Customers Like You: summary for Sarah

To: Sarah Okafor, Head of E-Commerce **From:** Data & ML Team (Jagadeesh Aravind Kalyan Shetti) **Date:** 13 October 2026

The short answer

We built a small, fast "customers who bought this also bought" engine, and it runs in about 2.4 seconds. But on purchases it never saw, it **did not beat your best-seller list**: it was right for about 21 in 100 customers, the best-seller list for about 27 in 100. **Our advice: do not switch the product pages to it yet** (next steps below). We also found three customer groups you can name, but they mostly differ in how much people buy.

Which method, and why

We chose **Truncated SVD**. It boils the customer-by-product table down to 200 "shopping patterns", and each product gets a short fingerprint. Two products with similar fingerprints are bought by the same customers, which is exactly what the box needs. We picked it because the table is 97.95% empty (5,473 customers x 3,687 products) and SVD works on it as it is. We also ran the rival method, PCA, with the same settings: it scored almost the same (see below), so the choice did not change the answer.

What we lose by compressing

With 200 patterns, about **64%** of the original table is not rebuilt (we measure the size of what is missing against the whole table). That number says how well we copy past buying, not how good the suggestions are, so we tested the suggestions directly.

How big and how fast

The full table would need about **161 MB**; the compressed version needs about **14.7 MB** (about 11 times smaller; stored sparse, the raw table is 5 MB). The whole nightly job (refit, the box for all 3,687 products, top 10 for all 5,473 customers) took about **2.4 seconds** on a laptop. One caution: our scoring step temporarily builds large tables of about 160 MB, so the job is not tiny in memory yet; splitting it into batches would fix that.

Does it beat best sellers?

We hid the last two months of purchases, then asked: for customers who bought something *new to them* in that period, did at least one of our 10 suggestions match? That is the "hit rate at 10". Test: 1,913 customers, 44,727 purchases the model never saw.

Method	Hit rate at 10	Meaning
Best sellers (10 top products they have not bought)	**0.2687**	about 27 in 100 customers
Truncated SVD (our model)	0.2148	about 21 in 100 customers
PCA (rival method, same settings)	0.2180	about 22 in 100; gap to SVD is about 6 customers

Best sellers won by about 5.4 percentage points. This is one split and one run, so it is a measurement, not a proven gap, and we did not change our settings after seeing it. A likely reason (not yet tested): popular products are bought by many people, while suggestions built from one customer's history lean towards niche items.

Customer groups for marketing

Group (our name)	Customers	Distinct products each	Main countries
Occasional small buyers	4,748 (87%)	48	UK 92%, Germany 1%, France 1%
Regular core buyers	712 (13%)	238	UK 88%, Germany 4%, France 2%
Largest accounts	13 (0.2%)	1,146	UK 69%, EIRE 15%, Australia 8% (of 13 customers)

Honest caveat: the groups are mostly **volume tiers** (small, regular, very large), not different tastes. The core group leans to retro-spot cake cases, lunch bags and jumbo bags. We did not call anyone "wholesale", because the data does not show who is. No customer IDs appear in this memo or in the map.

What could go wrong

- **New customers.** About 22% of new purchases (13,829 of 63,252) came from customers with no history. The engine cannot serve them; show best sellers.
- **Rare products.** We dropped products with fewer than 10 buyers: 19% of products (842 of 4,529), so they get no suggestions.
- **Repeat buying.** About 39% of order lines were repeats of something already bought, and 33% of last-two-months purchases (31,766 of 95,018) were repeats we excluded. A reorder reminder is a different feature.
- **Near-duplicates.** Colour or size variants of one product can fill a whole suggestion list.
- **Thin evidence.** One time split and one run; the largest-accounts group is only 13 customers.

Next steps

(1) Pick the number of patterns properly with a validation slice of older data. (2) Try a blend that mixes best sellers with personal suggestions. (3) Split the scoring into batches to cut memory. (4) Show best sellers to new customers. (5) Consider reorder reminders.